

- Q.31 What are the design parameters of open wells?
- Q.32 What are physical properties of soil influencing irrigation?
- Q.33 State the volumetric method of water measurement.
- Q.34 Define aquifer constant and specific capacity of wells.
- Q.35 Describe the working of submersible pumps.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What are different surface methods of irrigation. Explain the basic details, characteristics, type and adaptability of Border irrigation method.
- Q.37 Explain the constructional detail of centrifugal pump along with its principle of operation. Also state common trouble and remedies.
- Q.38 Describe the sprinkler irrigation system along with its components. Also describe its adaptability and layout diagram.

No. of Printed Pages : 4

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Roll No.

3rd Sem / Agri Engg. Subject:- Farm Irrigation Engg.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The number of days that can be allowed for applying one irrigation to a given design area during the peak consumptive use period of crop being irrigated
- a) Duty b) Irrigation period
- c) Irrigation frequency d) Delta
- Q.2 The rate at which water infiltrates into the soil.
- a) Infiltration rate b) Irrigation Frequency
- c) Gross irrigation d) Delta
- Q.3 The downward movement of water through the soil is the
- a) Infiltration b) Absorption
- c) Porosity d) None of the above
- Q.4 The height at which water stand in a piezometer
- a) Static water level b) Piezometric surface
- c) pumping water level d) None of these
- Q.5 The ratio of power output to power input
- a) Water horse power b) Shaft horse power

- c) Pump Efficiency d) Brake horse power
- Q.6 The total amount of water applied through irrigation is termed as
- a) Net irrigation requirement
b) Gross irrigation requirement
c) Duty
d) Delta
- Q.7 The total depth of water application to the crop is called:
- a) Duty b) Delta
c) Base d) None of these
- Q.8 The line connecting points of equal elevation
- a) Contour b) Isobath
c) Isobar d) Drawdown
- Q.9 The actual area irrigated in a year from outlet is called
- a) Irrigation period b) Intensity of irrigation
c) Irrigation frequency d) None of these
- Q.10 Which irrigation method is also called as trickle irrigation?
- a) Sprinkler Irrigation b) Furrow irrigation
c) Drip Irrigation d) Check Flooding

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 State two methods of surface irrigation.

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- Q.12 Give two units of water measurement.
- Q.13 _____ irrigation method is also called as trickle irrigation?
- Q.14 Give the purpose of canal lining.
- Q.15 Give the use of priming.
- Q.16 List two limitations of drip irrigation system.
- Q.17 What is the necessity of irrigation?
- Q.18 Give two physical properties of soil.
- Q.19 Define water distribution efficiency.
- Q.20 State two design elements of tubewells.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are different factors affecting Duty?
- Q.22 What are different type of impellers used in centrifugal pump?
- Q.23 Describe confined and unconfined aquifers.
- Q.24 State different methods of ground water recharge.
- Q.25 State four characteristics of aquifer influencing yield of well.
- Q.26 Describe the working of drip irrigation system.
- Q.27 What are ill effects of irrigation.
- Q.28 What are the advantages of lining canals?
- Q.29 Write down the criteria for selection of irrigation pumps.
- Q.30 What is the need and importance of irrigation?

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